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SUPERSEDING
MIL-C-915/69(SHIPS)
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MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, THREE CONDUCTOR, TYPE TNW

(A) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, uncoated. (See details for size.)
Second - Separator, polyester type G of MIL-I-631, opaque white, applied either helically or longitudinally to give 100 percent coverage.
(A) Third - Cross-linked thermosetting polyethylene insulation, colored black. See details for thickness. Standard Identification Code applied by Method 1.
Fourth - Three conductors cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm, well-rounded assembly.
Fifth - Binder tape of 0.002 inch thick, corrugated polyester, type G of MIL-I-631.
Sixth - Standard thermoplastic jacket (gray). (See details for thickness.) Cable surface marking required.

Details

Type and size	Conductor Navy Standard size	Insulation thick. Nominal (inch)	Cable jacket thick. nominal (inch)	Overall diameter maximum (inches)	Conductor resistance per 1000 ft. at 25°C. maximum (ohms)	Insulation resistance/ 1000 ft. minimum (megohms)	Voltage withstand minimum (volts)	Cold bending mandrel diameter maximum (inches)	Accelerated service loading current (amperes)
TNW-3	3(7)	0.030	0.040	0.411	4.00	6000	4000	3	---
TNW-4	4(7)	.030	.040	.449	2.51	5000	4000	4	---
TNW-9	9(7)	.030	.050	.625	1.25	3800	4000	5	---
TNW-14	14(7)	.045	.050	.670	.785	4400	5500	5	90
TNW-23	23(7)	.045	.055	.760	.496	3700	5500	7	115
TNW-50	50(19)	.045	.060	.969	.23	2600	5500	9	185
TNW-75	75(37)	.055	.070	1.134	.149	2600	7000	10	250
TNW-100	100(61)	.055	.075	1.266	.114	2400	7000	12	295
TNW-150	150(61)	.055	.075	1.515	.0720	1800	7000	14	400

(A) Denotes changes

FSC 6145

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EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - Ohms/1000 ft., maximum	See details
Voltage withstand - Volts, rms, minimum	See details
Insulation resistance - Megohms/1000 ft., minimum	See details
Group A:	
Visual and dimensional	
Group B:	
Cold bending, cable - (See details for mandrel diameter)	No failure
Physicals (unaged)	
Insulation	
Tensile strength - Psi, minimum	1800
Elongation - Percent, minimum	250
Jacket	
Tensile strength - Psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Permanence of printing (conductor) - Cycles, minimum	25
Permanence of printing (jacket) - Cycles, minimum	250
Cable filler removability	2
Flammability - Inches, maximum	30
Heat distortion - percent of unaged maximum	Zero
(A) Drip - At 95° ± 1°C.	No failure
(A) Accelerated service - (Sizes 14 and larger)	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - Percent of unaged, minimum	75
Elongation - Percent of unaged, minimum	75
Jacket	
Hot oil immersion	
Tensile strength - Psi, minimum	1400
Elongation - Inches, minimum	1

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C plus the following:

Cold working (minus 20° + 2°C.)	No damage
(A) Cable aging and compatibility (125° ± 5°C.)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
TNW-3	2500
TNW-4, 9, 14	2000
TNW-23	1500
TNW-50, 75	1000
TNW-100, 150	1000

(A) Custodians:
Army - MI
Navy - SH
Air Force - 80

(A) Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

(A) User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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